

Telink

Amazon Sidewalk

Quick Start Guide

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Keyword

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Brief

This document provides a quick start guide for the Amazon Sidewalk.

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Revision History

Version	Revisions
V1.0.0	Initial Release
V1.0.1	Modified ML3238S pin assignment; added BLE/FSK/LoRa switching function to Sidewalk Demo.
V1.1.0	Updated the entire document.

Telink Semiconductor

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1 Overview

Amazon Sidewalk is a secure community network protocol for the Internet of Things (IoT) developed by Amazon. For details about Telink's Amazon Sidewalk solutions and development resources, please visit the official Telink website: [Amazon Sidewalk](#).

This quick start guide helps users quickly set up the Amazon Sidewalk development environment, covering hardware and software preparation, tool installation, AWS configuration, firmware flashing and demo verification. For detailed development resources of the AIOT-DK1 + ML3238S Sidewalk development kit, please refer to the [Amazon Sidewalk User Manual](#).



Figure 1.1: Data Transmission Process

Related Documents & Tools

- [Amazon Sidewalk User Manual](#)
- [AIOT-DK1 User Manual](#)
- [Telink Burning and Debugging Tool](#)
 - [Burning and Debugging Tool \(Windows\) User Guide](#)
 - [Burning and Debugging Tool \(Linux & MAC\) User Guide](#)

2 Hardware Preparation

- Telink AIOT-DK1 development motherboard
- Telink ML3238S daughterboard (for Amazon Sidewalk)
- USB Type-C cable (for power supply and firmware downloading)
- Sidewalk Gateway (Echo 4 used in this demo)
 - Refer to the [List of Sidewalk Gateways](#) for more options.
- Wi-Fi router (Spitz AX (GL-X3000, Wi-Fi 6, VPN-supported, dual SIM))
 - The gateway needs Wi-Fi network access to connect to AWS Cloud.

3 Software Preparation

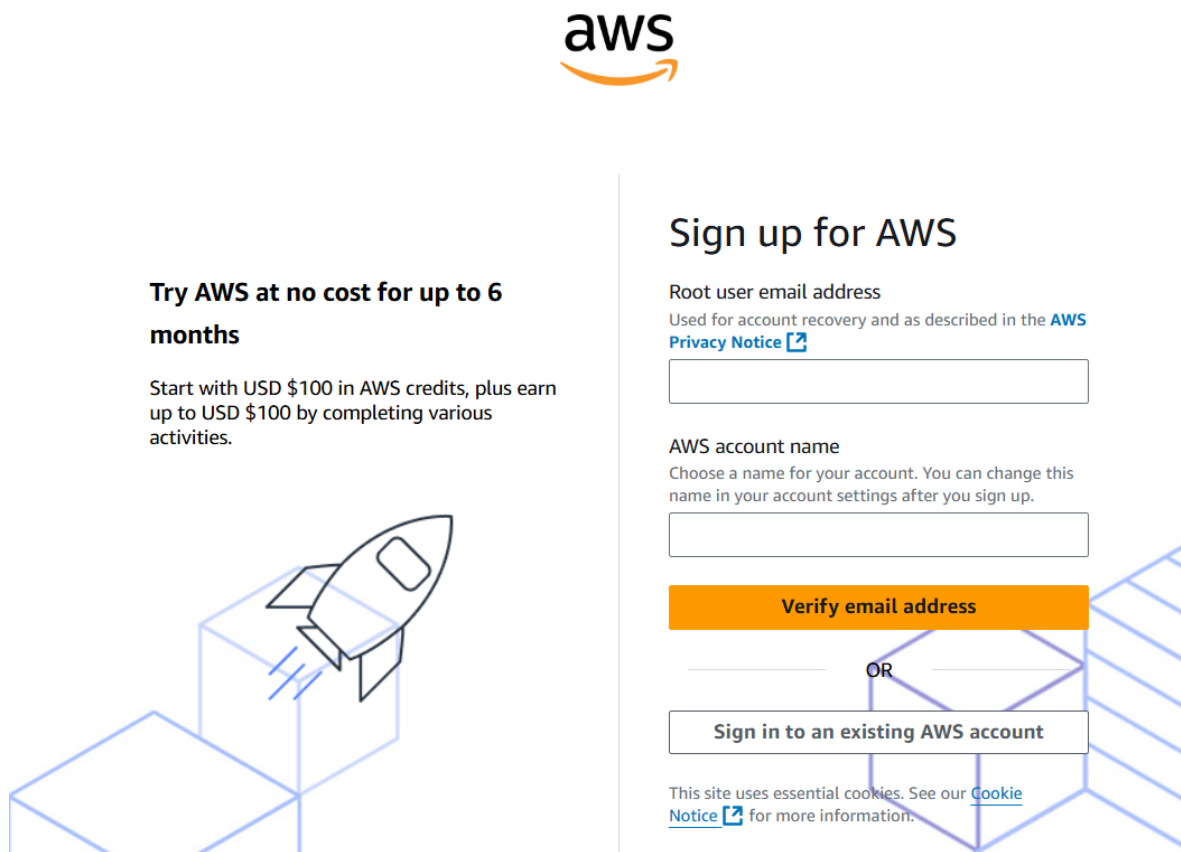
Two firmware files are required:

- **Test Firmware:** Compilable from the Telink Sidewalk SDK.
 - A precompiled version [sidewalk_sensor_monitoring.bin](#) is provided for quick test setup.
- **MFG Configuration Firmware:** Generate this file by following the instructions in the subsequent chapters.

4 Prepare MFG Configuration File

4.1 Set Up AWS Account

Create an AWS account via the official link: [Set up an AWS account](#).



The screenshot shows the AWS 'Sign up for AWS' page. On the left, there is a promotional banner for a 6-month free trial of AWS, mentioning a \$100 credit and a contest to win up to \$100. Below the text is a line drawing of a rocket launching from a cube. On the right, the 'Sign up for AWS' form is visible. It includes a 'Root user email address' field with a link to the AWS Privacy Notice, an 'AWS account name' field with instructions, a 'Verify email address' button, an 'OR' separator, and a 'Sign in to an existing AWS account' button. At the bottom, there is a cookie notice.

Figure 4.1: Set Up an AWS Account

After successful registration, log in to your AWS account.

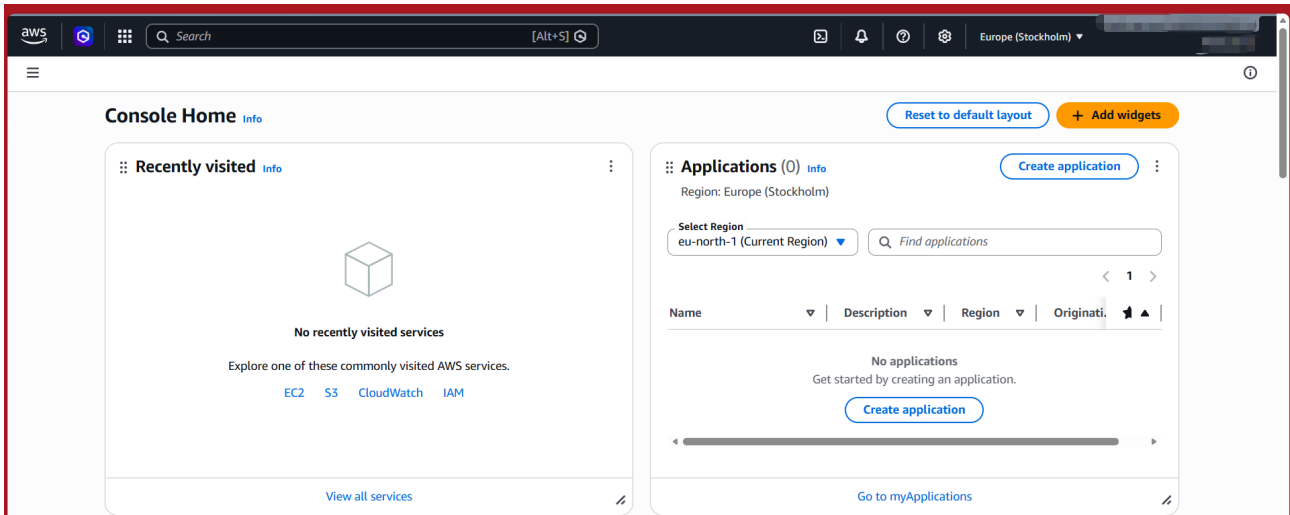


Figure 4.2: AWS Account Sign-In Page

4.2 Extract the Device Certificate

Reference: [What is AWS IoT Core?](#)

Open the IoT Core service in the AWS Management Console (use the search bar if it is not clearly visible in the UI).

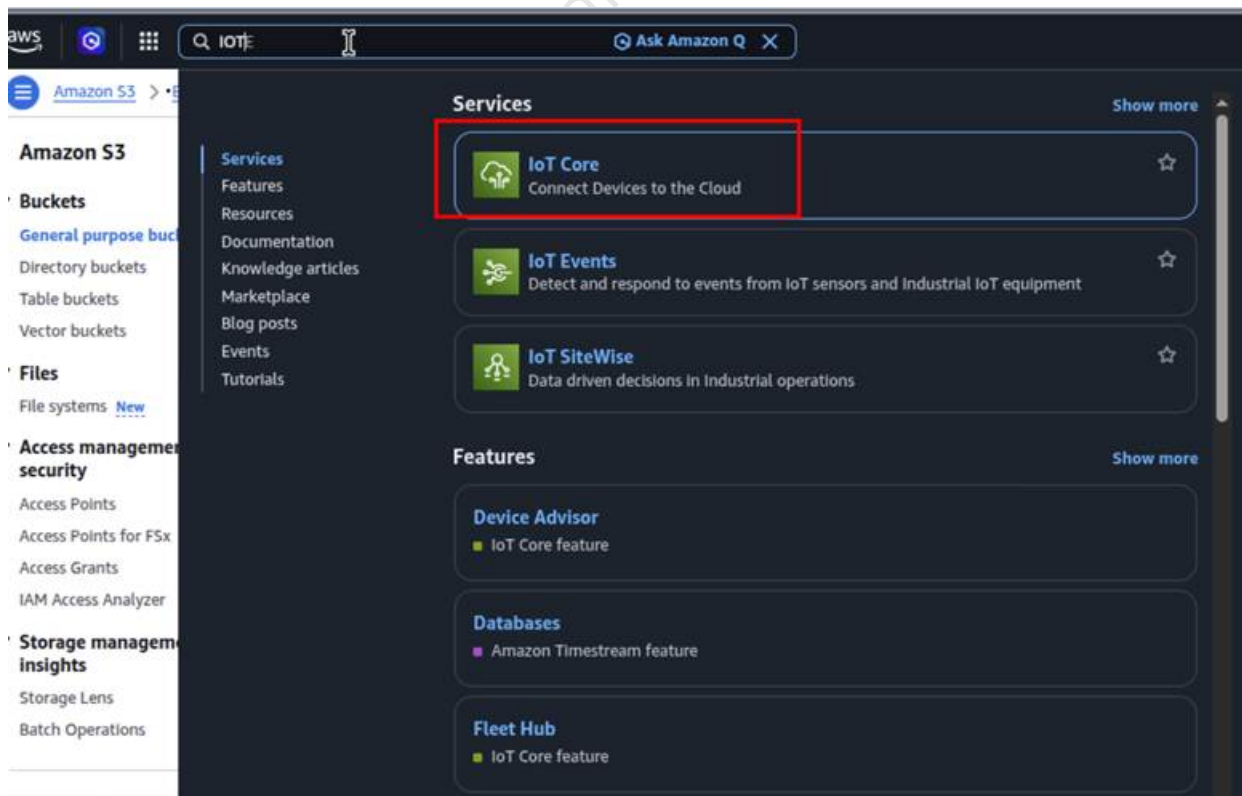


Figure 4.3: IoT Core

Navigate to your device (note: due to regional restrictions, Sidewalk devices may not appear in all regions).

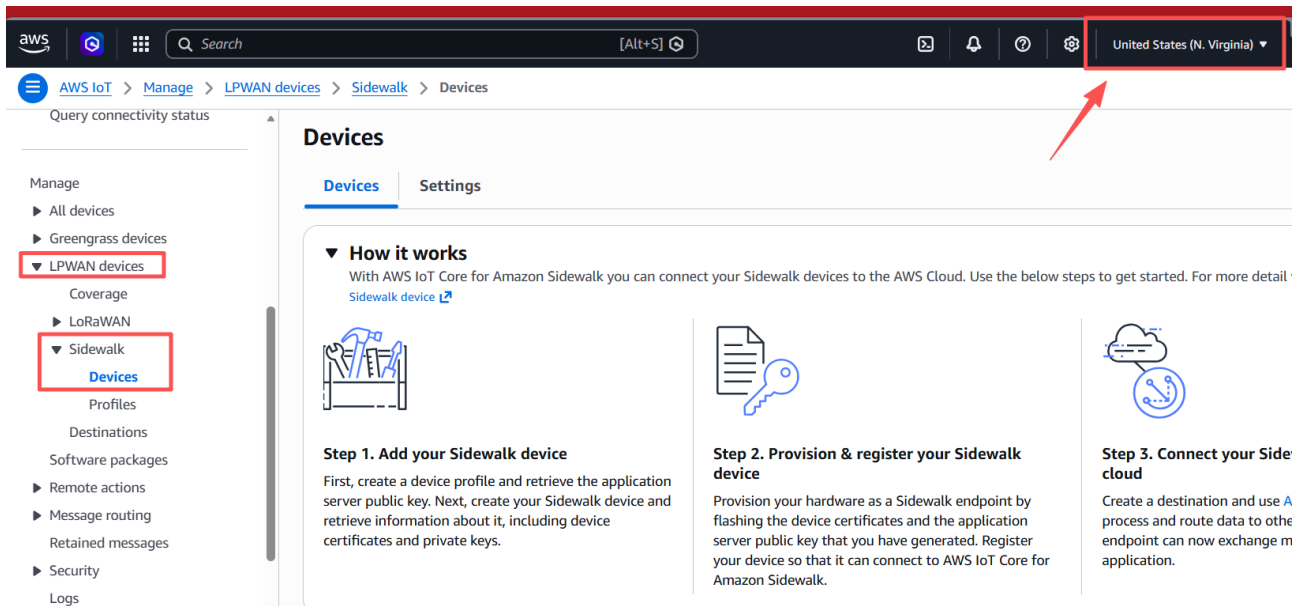


Figure 4.4: Sidewalk Device

Create a **Profile**:

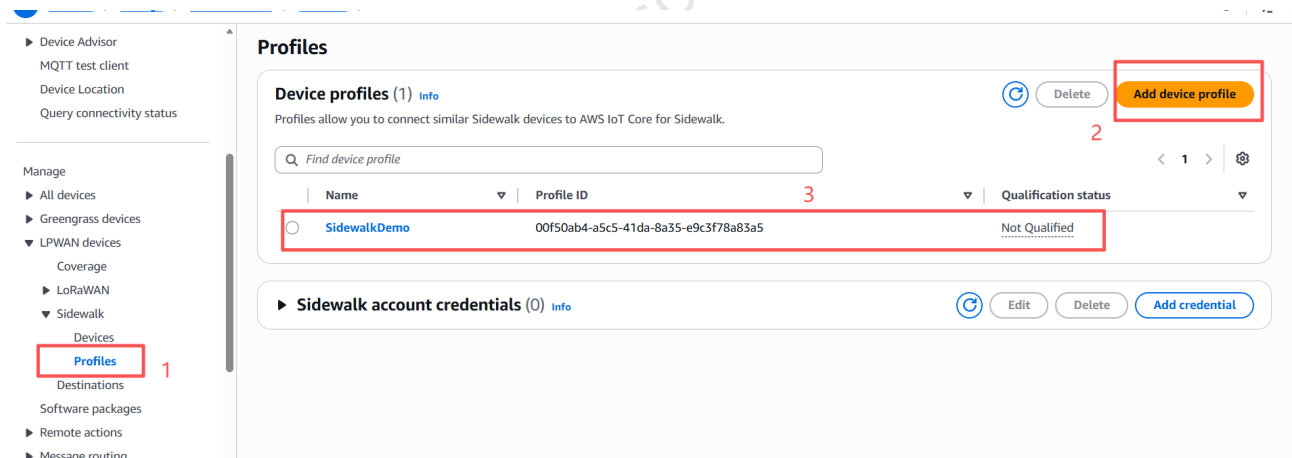


Figure 4.5: Create Profile

Create a **Destination**:

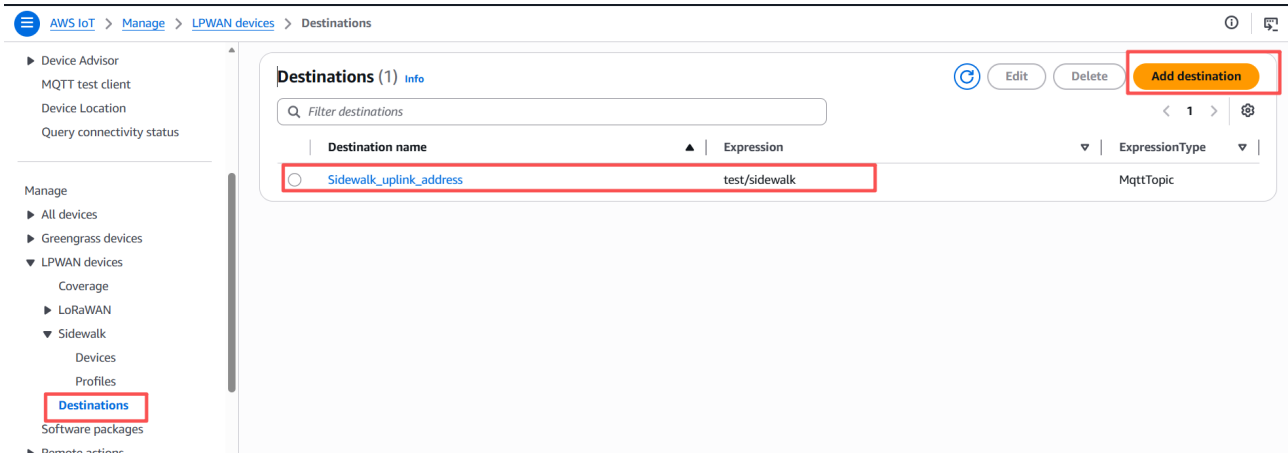


Figure 4.6: Create Destination

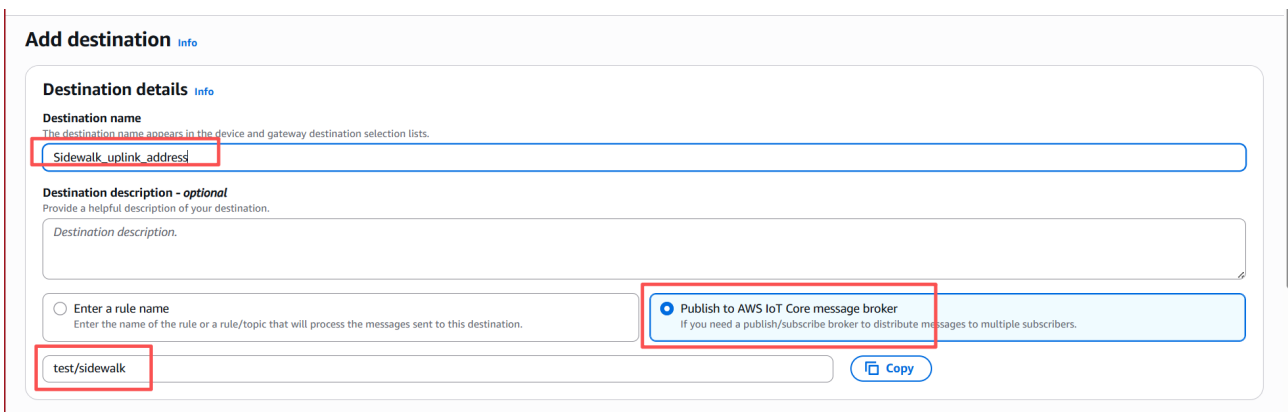


Figure 4.7: Destination Details

Create a **Sidewalk device**:

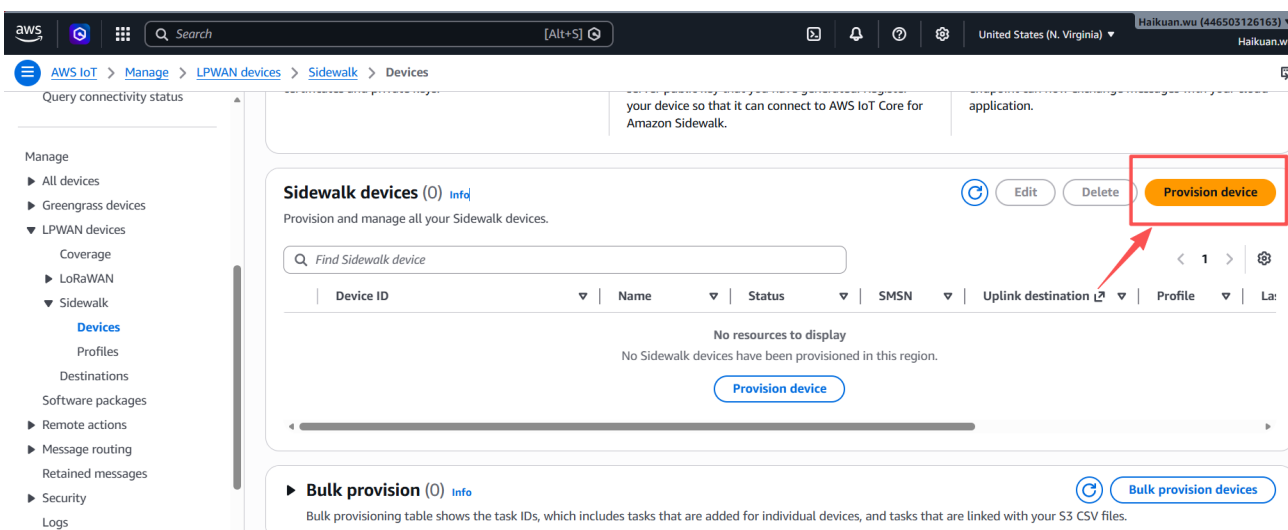
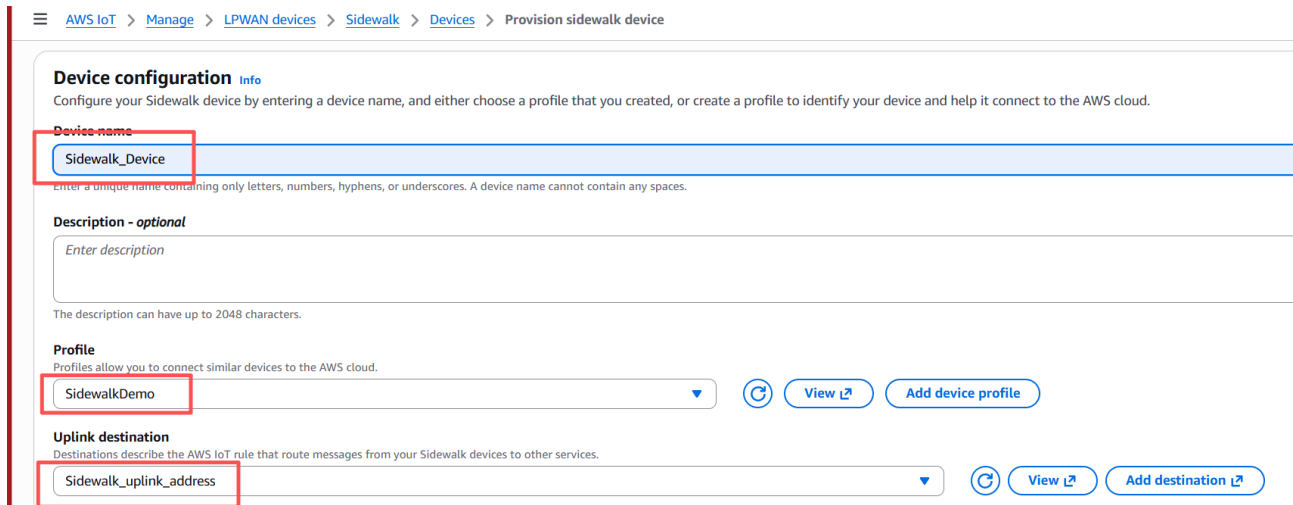


Figure 4.8: Create a Sidewalk Device

Select the profile and destination created earlier, finish device configuration and click **Create**.



Device configuration Info

Configure your Sidewalk device by entering a device name, and either choose a profile that you created, or create a profile to identify your device and help it connect to the AWS cloud.

Device name

Sidewalk_Device

Enter a unique name containing only letters, numbers, hyphens, or underscores. A device name cannot contain any spaces.

Description - optional

Enter description

The description can have up to 2048 characters.

Profile

Profiles allow you to connect similar devices to the AWS cloud.

SidewalkDemo

[View](#) [Add device profile](#)

Uplink destination

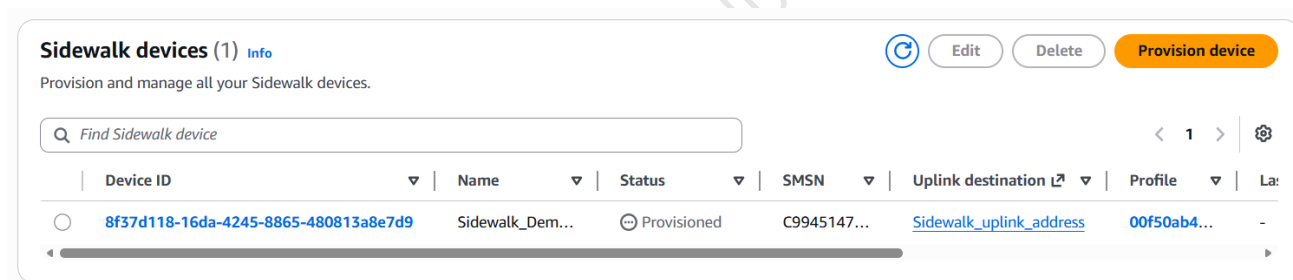
Destinations describe the AWS IoT rule that route messages from your Sidewalk devices to other services.

Sidewalk_uplink_address

[View](#) [Add destination](#)

Figure 4.9: Device Configuration

Click the created device, then download its JSON configuration file.



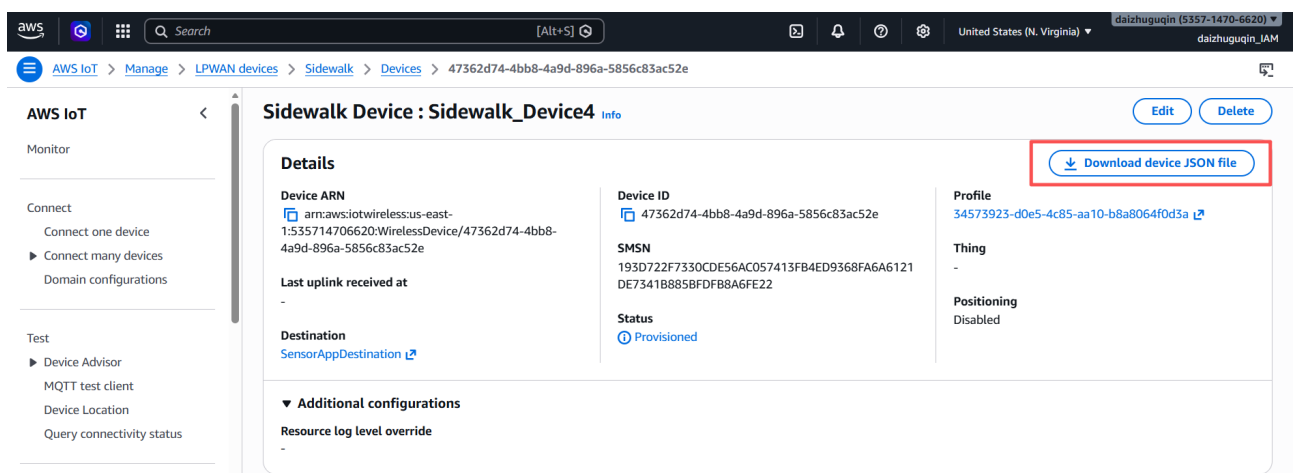
Sidewalk devices (1) Info

Provision and manage all your Sidewalk devices.

[Find Sidewalk device](#)

Device ID	Name	Status	SMSN	Uplink destination	Profile	La
8f37d118-16da-4245-8865-480813a8e7d9	Sidewalk_Dem...	Provisioned	C9945147...	Sidewalk_uplink_address	00f50ab4...	-

Figure 4.10: Click the Created Device



Sidewalk Device : Sidewalk_Device4 Info

[Edit](#) [Delete](#) [Download device JSON file](#)

Details

Device ARN
arn:aws:iotwireless:us-east-1:535714706620:WirelessDevice/47362d74-4bb8-4a9d-896a-5856c83ac52e

Device ID
47362d74-4bb8-4a9d-896a-5856c83ac52e

SMSN
193D722F7330CDE56AC057413FB4ED9368FA6A6121
DE7341B885BFD8A6FE22

Status
Provisioned

Profile
34573923-d0e5-4c85-aa10-b8a8064f0d3a

Thing
-

Positioning
Disabled

Additional configurations

Resource log level override
-

Figure 4.11: Download Device JSON File

4.3 Generate the MFG Binary

Get the Telink Sidewalk SDK: https://github.com/telink-semi/tl_amazon_sidewalk.

Navigate to the provisioning tool directory in the Telink Sidewalk SDK:

```
./tl_amazon_sidewalk/tl_sidewalk_sdk/vendor/Amazon_sidewalk/sidewalk/apps/common/sidewalk_sdk/
tools/provision/
```

Copy the certificate.json file (obtained in the section [Extract the Device Certificate](#)) to this directory.

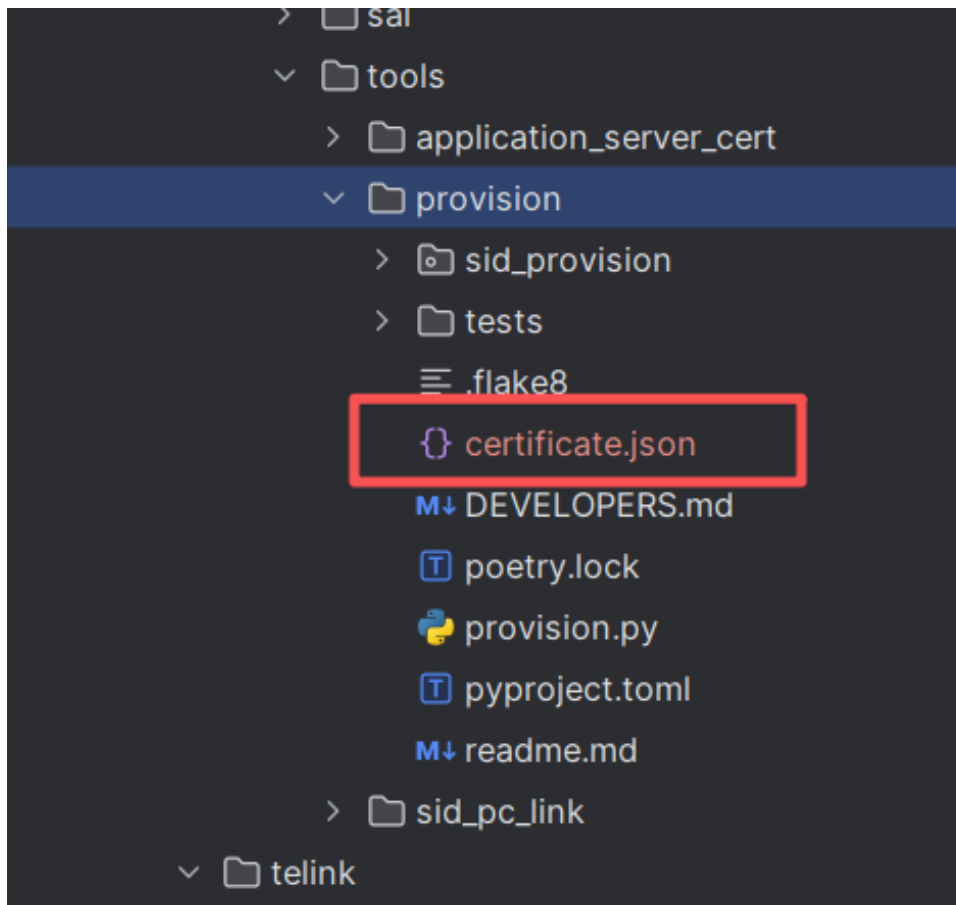


Figure 4.12: Copy certificate.json File

Run the provision.py script (Python 3 required) to generate the MFG binary:

```
python provision.py telink aws --output_bin mfg.bin --certificate_json certificate.json
```

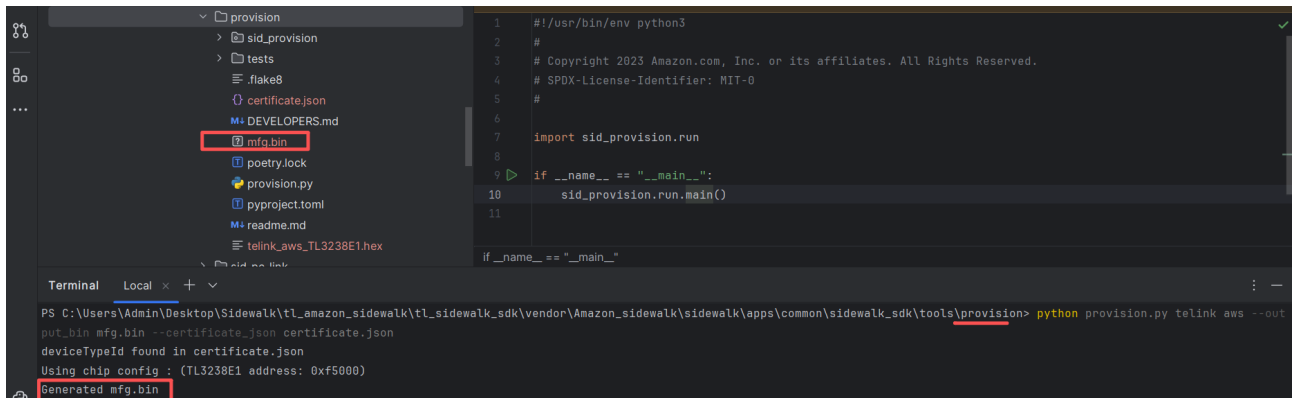


Figure 4.13: Run the `provision.py`

Note:

- For additional details on MFG file generation, refer to the [Amazon Sidewalk User Manual](#).

5 Test the Sidewalk Demo

Prepare the required hardware, power on the Wi-Fi router, and the Sidewalk Gateway. Add the gateway to the Wi-Fi network via the [Alexa app](#) to establish a connection with the AWS cloud.



Figure 5.1: Test the Sidewalk Demo

5.1 Flash the Test Firmware

Flash the [sidewalk_sensor_monitoring.bin](#) to address **0x00000000**.

Flash the MFG.bin (generated in the previous steps) to address **0x000f5000**.

Firmware Burning Procedure:

- Power on the device and connect it to the PC via USB cable as shown below:

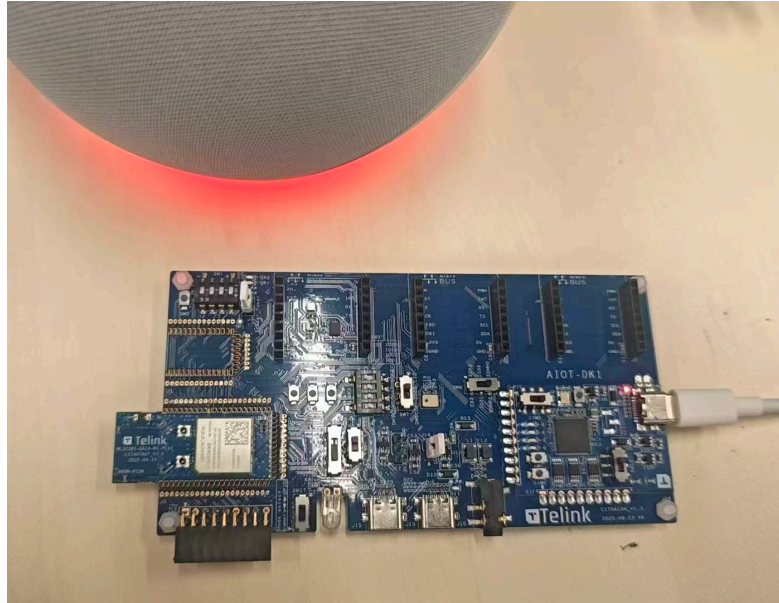


Figure 5.2: Power On the Device

- Download [Telink BDT](#). Refer to the documents below for full burning instructions:
 - [Burning and Debugging Tool \(Windows\) User Guide](#)
 - [Burning and Debugging Tool \(Linux & MAC\) User Guide](#)

Burning and Debugging Tool

The Burning and Debugging Tool is applicable to the application development of Telink series SoC. During the SDK development process, the firmware can be directly downloaded to the target development board via USB mode or Burning EVK. The main functions include: erasing Flash sectors, firmware download, activating the MCU, accessing memory Spaces including Flash/Core/Analog/OTP, reading/writing global variables and viewing USB logs.



It supports main operating systems such as Windows, MAC and Linux, and provides command-line tool operation modes.



It has multiple testing tools such as BQB, EMI, and Secure Boot, offering rich and powerful functions.



It integrates programming and debugging functions, supporting multiple connection methods such as SW5 and USB.

Download the latest version of Burning and Debugging Tool

Windows 

Linux 

MAC Intel 

Mac AArch64 

Burning and Debugging Tool User Guide (Windows) 

Burning and Debugging Tool User Guide (Linux&MAC) 

Figure 5.3: Burning and Debugging Tool

- Launch the BDT tool and select **On-board Programmer**.



Figure 5.4: On-board Programmer

- Follow the steps shown below to complete firmware flashing.

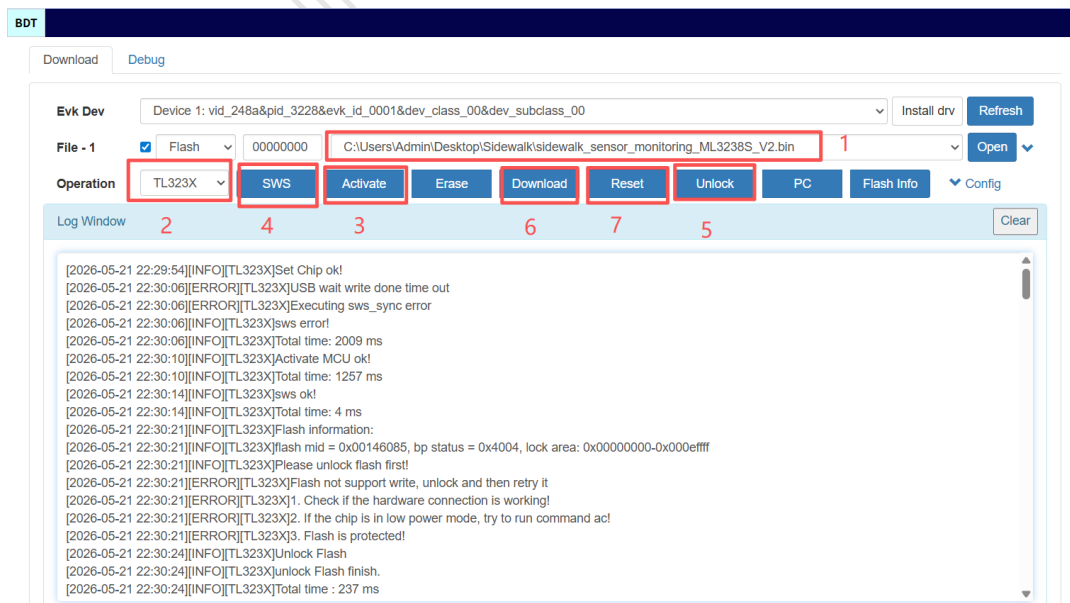


Figure 5.5: Flashing firmware

In the figure above:

- "1": Select the firmware to be programmed.
- "2": Select the chip model.
- "3": Activate the chip (if the chip has entered low-power mode, it can be woken up).
- "4": Confirm successful SWS communication.
- "5": Unlock the flash.
- "6": Download the firmware.
- "7": Reset the chip (software reset).

5.2 Verify Connection Between DUT and Gateway

LED1 and LED2 remain on after flashing.

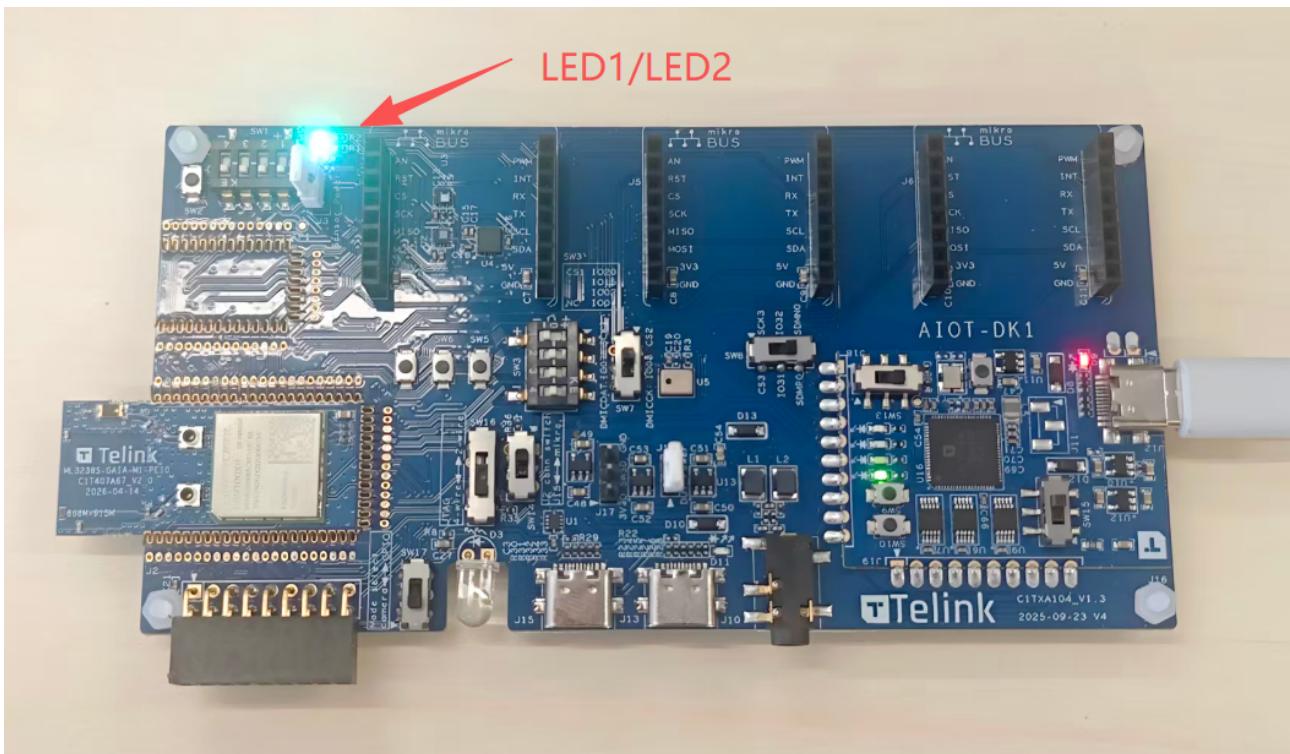


Figure 5.6: Hardware Status

The DUT advertises as SID_APP and can be detected in nRF Connect. Do not connect to it using a mobile phone.

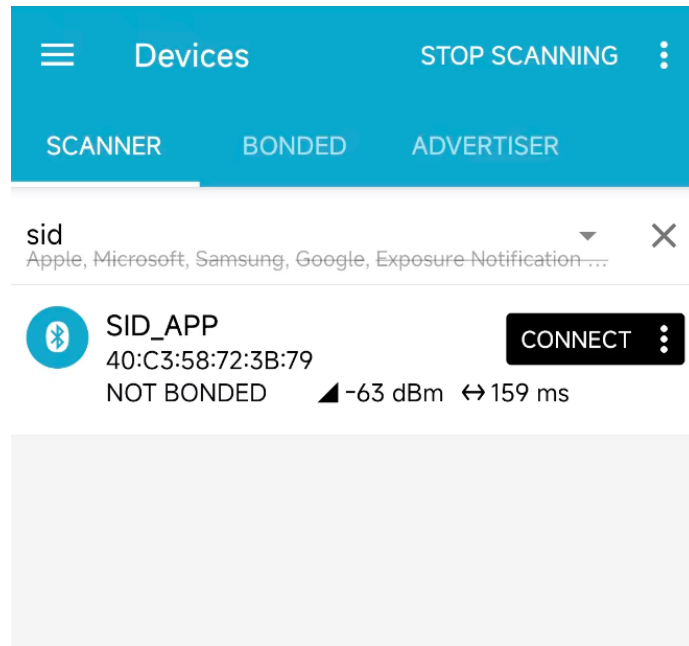


Figure 5.7: DUT Advertise

The DUT automatically searches for and connects to a nearby gateway. Once connected, both LEDs turn off, and SID_APP is no longer visible because the device stops advertising.

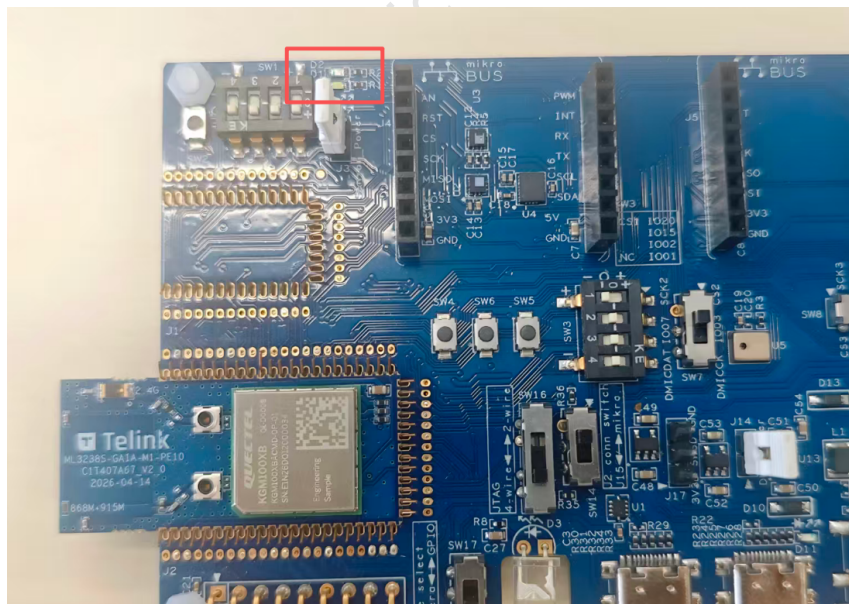


Figure 5.8: Hardware Status

5.3 Open the AWS IAM Console

After flashing completes, LED1 and LED2 stay steadily on. Both LEDs turn off simultaneously once the DUT connects to the Sidewalk gateway. Then open the AWS IAM Console to check the data reporting status.

Log in to your AWS account and access AWS IoT -> MQTT test client.

Subscribe to the topic with `test/sidewalk`, which was defined when creating the destination. Alternatively, use the wildcard `#` to subscribe to all messages.

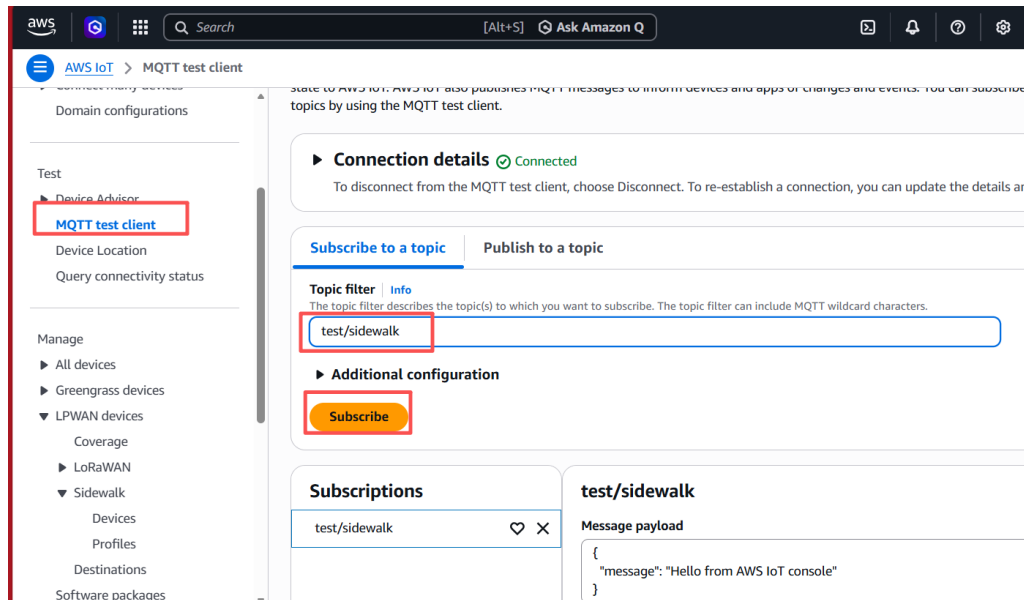


Figure 5.9: MQTT Test Client

Real-time data transmission from the DUT confirms successful environment deployment.



Figure 5.10: MQTT Test Log

Congratulations, you have completed the basic environment setup. Refer to the [Amazon Sidewalk User Manual](#) for more Telink Sidewalk-related information.